

Fourfold forrelation at $N = 4096$: a parallel-probe certificate

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Abstract

Three single photons, each crossing two phase masks, decide the fourfold-forrelation promise with hard dose six and error $81/256$. The recorded finite certificate excludes a single parallel probe of dose six whose state is block diagonal in total signal photon number. Its comparison is between a two-pass interferometer and one parallel interrogation. A previously claimed extension to classically adaptive fresh probes has a norm-interface gap and is not established here. We give the short analytic argument, the exact interface to the numerical certificate, and the remaining proof obligation.

1 The comparison

For a power of two N , let $H_{ij} = N^{-1/2}(-1)^{i \cdot j}$ and $u = N^{-1/2}(1, \dots, 1)^\top$. Four unknown binary phase masks $D_b = \text{diag}(x^{(b)})$, $x^{(b)} \in \{\pm 1\}^N$, define

$$F(x) = \text{Forr}_4(x) = u^\top D_1 H D_2 H D_3 H D_4 u. \quad (1)$$

Decide $F \geq 1/4$ versus $F \leq -1/4$, with worst-case error at most $1/3$. This is a variant of the query problem of Aaronson and Ambainis [1]; a classical-query lower bound is not a bound on the quantum probes below.

A parallel probe is fixed before observing any sample output, undergoes one direct-sum mask interaction, and is measured by an arbitrary POVM. It may have arbitrary bypass idlers, repeated signal modes, entanglement, and coherence within a fixed total signal-number sector. We require

$$[\rho, \hat{N}_{\text{sig}}] = 0, \quad \rho \text{ supported on } \hat{N}_{\text{sig}} \leq 6. \quad (2)$$

Thus mixtures of different total numbers are allowed, but coherence between them is not. Dose counts every photon-mask traversal; it is a support bound, not a mean or a postselected cost. Several predetermined batches of total dose six also fit this model by joint preparation and a final collective measurement.

Theorem 1 (Finite parallel comparison). *At $N = 4096$, the recorded occurrence certificate gives two laws P_+, P_- supported on the respective promises such that every parallel probe satisfying (2) has output laws obeying*

$$\text{TV}(T_+, T_-) \leq 0.260969224792207925. \quad (3)$$

Its equal-prior Bayes error is therefore at least $0.369515387603896037 > 1/3$. A multipass protocol has hard dose six and error $81/256 < 1/3$.

The analytic proof reduces the lower bound to explicit occurrence coefficients stored with their supporting derivations. Section 5 specifies this interface and its arithmetic verification. The package was developed with substantial language-model assistance. The coefficient families and their reduction to physical occupations remain subject to independent mathematical review; successful replay checks the recorded calculation rather than every analytic lemma.

2 The two-pass protocol

Define

$$L_x = D_2 H D_1 u, \quad R_x = H D_3 H D_4 u.$$

Prepare a single photon in $(|0\rangle L_x + |1\rangle R_x)/\sqrt{2}$. Each arm crosses exactly two masks. Recombining the paths gives

$$\Pr(X = \pm 1 \mid x) = \frac{1 \pm \langle L_x, R_x \rangle}{2} = \frac{1 \pm F(x)}{2}. \quad (4)$$

The mode label can be discarded; only the output port is needed. At either promise boundary, a flag is correct with probability $5/8$. Three independent flags and majority decoding have error

$$(3/8)^3 + 3(5/8)(3/8)^2 = 81/256,$$

and deterministic dose $3 \cdot 2 = 6$. No outcome is discarded. This proves the upper bound for every promised input and every N .

3 The hard inputs

Set $q = 64$, so $N = q^2$. Identify a coordinate with $(a, b) \in [q]^2$, and use row-major flattening, so $H_N = H_q \otimes H_q$. Write $S_q = \sqrt{q} H_q$. For a uniform permutation π and independent uniform signs σ_b , define arrays

$$L_{a,b} = S_q(a, \pi(b))\sigma_b, \quad R_{\pi(b),a} = \sigma_b S_q(b, a). \quad (5)$$

Their useful property is simply

$$H_N L = R, \quad H_N R = L. \quad (6)$$

Indeed, at coordinate (c, d) ,

$$(H_N L)_{c,d} = \frac{1}{q} \sum_{a,b} S_q(c, a) S_q(d, b) S_q(a, \pi(b)) \sigma_b = S_q(d, b_c) \sigma_{b_c} = R_{c,d},$$

where $b_c = \pi^{-1}(c)$. The second identity follows from $H_N^2 = I$.

Take three independent pairs (L_j, R_j) and set

$$z = (L_1, R_1 \odot L_2, R_2 \odot L_3, R_3). \quad (7)$$

Starting on the right of (1), (6) successively cancels $R_3, L_3, R_2, L_2, R_1, L_1$ and leaves u . Thus $F(z) = 1$ exactly. Negating the first block changes the value to -1 .

Now independently flip every coordinate with probability $3/25$. Equivalently, multiply it by a sign η with mean

$$\beta = 19/25.$$

Let $\tilde{P}_+$ be the resulting positive law and $\tilde{P}_-$ its first-block reflection. Then $\mathbb{E}_{\tilde{P}_+} F = \beta^4$. Finally condition each law onto its own promise to obtain P_+, P_- .

Lemma 2 (Cost of conditioning). *The conditioning loss is at most 0.002225128406630703.*

Proof. Fix an exact plant z . Reveal the four noisy blocks in order from left to right and form the Doob martingale of F . When block b is exposed, the future $4 - b$ masks contribute their means, giving a factor β^{4-b} . By (6), the remaining deterministic right-hand chain is a sign vector divided

by $\sqrt{N}$. The left-hand chain has norm one. The sum of squared coefficients of the newly exposed signs is therefore $\beta^{2(4-b)}/N$.

A sign in $[-1, 1]$ has centered subgaussian proxy at most one by Hoeffding's lemma. Conditional independence within each newly exposed block gives the martingale proxy

$$V_\beta = \frac{1 + \beta^2 + \beta^4 + \beta^6}{N} = \frac{(1 + \beta^2)(1 + \beta^4)}{N}.$$

With $g = \beta^4 - 1/4$ and $x = g^2/(2V_\beta)$, Chernoff gives a failure probability at most e^{-x} . This holds for every plant. Reflection gives the same bound on the negative side. The verifier evaluates the exact rational upper

$$2e^{-x} \leq 2(1 + x/4096)^{-4096} < 0.002225128406630703.$$

□

4 From a quantum probe to one matrix

This is where the single-batch hypothesis is used. Purify the input and write $\psi = \sum_\nu |\nu\rangle \alpha_\nu$, where ν is a physical Fock occupation, including repeated modes. Put $p_\nu = \|\alpha_\nu\|^2$ and let $\chi_\nu(x) = \prod_{b,i} (x_i^{(b)})^{\nu_{b,i}}$. For the unconditioned hard laws, define the half-difference moment matrix

$$K_{\nu\mu} = \frac{1}{2}(\mathbb{E}_{\tilde{P}_+} - \mathbb{E}_{\tilde{P}_-})[\chi_\nu \chi_\mu]. \quad (8)$$

The map $|\nu\rangle \mapsto |\nu\rangle \alpha_\nu / \sqrt{p_\nu}$ is an isometry on nonzero weights. Therefore the trace norm of half the difference of the averaged quantum output states is exactly

$$\|\text{diag}(\sqrt{p})K\text{diag}(\sqrt{p})\|_1. \quad (9)$$

Any subsequent measurement has total variation no larger than (9). Mixed inputs are handled by convexity and sector refinement. This proves the reduction for arbitrary idlers and POVMs.

The independent permutation signs make (8) vanish unless the symmetric-difference degree is odd in every block. A surviving profile $a = (a_1, \dots, a_4)$ therefore has degree four through twelve. Let t_b count its ket occurrences in block b . If n, m are the four block totals of the physical occupations, then

$$m = n + a - 2t, \quad 2|t| = |a|. \quad (10)$$

The second equality follows from equal total signal number in (2). Physical mode positions and distinctness constraints remain in the occurrence kernel $\mathcal{O}_{a,t}$; n, m, a, t are only four-component counts.

5 The finite coefficient certificate

For each contributing profile and split, the repository supplies a coefficient $c_{a,t}$ for the complete physical occurrence kernel:

$$\sup_{p,w} \|\text{diag}(\sqrt{p})\mathcal{O}_{a,t}\text{diag}(\sqrt{w})\|_1 \leq c_{a,t}. \quad (11)$$

Here p, w are arbitrary normalized row and column laws. The kernels include repeated-mode reduction, simple supports, and every distinctness mask crossing the row-column split. Discarding those masks was an error in an earlier coefficient proof.

The supporting derivations are retained in `code/source_snapshot/`. The entry point `code/COEFFICIENTS.md` maps the accepted families to their proof notes and executable formulas. The 888 balanced high-sector entries supplement the inherited lower-degree bounds. Of 6,016 high-sector routing entries, 5,128 are unbalanced and are excluded rather than used as coefficients. There are 272 unbalanced split/state incidences, forming 136 undirected edges in the larger cross-number space; they are a genuine missing case for number coherence.

Here is the complete interface between (11) and the numerical answer. Let

$$\mathcal{N} = \{n \in \mathbb{Z}_{\geq 0}^4 : |n| \leq 6\}, \quad |\mathcal{N}| = \binom{10}{4} = 210, \quad b(n, t) = \prod_{j=1}^4 \binom{n_j}{t_j}.$$

For every balanced (a, t) and compatible (n, m) in (10), add

$$\beta^{|a|} \widehat{c}_{a,t} \sqrt{b(n, t) b(m, a - t)} \quad (12)$$

to A_{nm} , and put $B = (A + A^\top)/2$. The factor $\beta^{|a|}$ is the attenuation of the surviving signs. The binomial factors count ways to select their photon occurrences.

The occurrence expansion, the trace-norm triangle inequality and (11) bound (9) by $v^\top B v$, where v_n is the square root of the probe's probability of block occupation n . Since $\|v\|_2 = 1$, this is at most $\rho(B)$, the Perron root. The matrix is nonnegative, symmetric and block diagonal in total photon number. This is a single-batch statement; no adaptive normalization has been inserted.

For nonzero binary64 source coefficients c , the numerical code uses $\widehat{c} = (\lceil 10^9 c \rceil + 1)/10^9$; zero coefficients stay zero. This rounds the supplied floating values outward. Analytic bounds must still be justified by the family derivations; rounding alone does not turn a floating formula into an analytic certificate. The dual-endpoint family additionally checks exact squared bounds.

A floating eigensolve provides only a strictly positive candidate w . The accepted bound is the outward Collatz ratio

$$\rho(B) \leq \max_i \frac{(\widehat{B}w)_i}{w_i} < 0.258744096385577223.$$

All entries and ratios are reconstructed with directed decimal arithmetic and outward square roots. The committed candidate makes replay independent of eigensolver rounding. Combining this with Lemma 2 gives (3). The stored high-precision sum is rounded once; adding the two displayed component bounds and rounding again would give a slightly looser value. The Bayes identity $(1 - \text{TV})/2$ completes Theorem 1.

6 The adaptive gap

For a fresh adaptive tree, the earlier proof constructs strategy coefficients

$$D(I, J) = \langle V_J, U_I \rangle, \quad \sum_I \|U_I\|^2 \leq 1, \quad \sum_J \|V_J\|^2 \leq 1.$$

This induction is valid. It does not justify applying (11) to the *sum of all entries*

$$\sum_{I, J} K_{f(I), f(J)} D(I, J).$$

The obstruction is elementary. For n histories take $K = J_n$ (the all-ones matrix) and scalar features $U_i = V_i = 1/\sqrt{n}$. Both squared feature masses are one and the kernel K has unit scalar Schur factors. Yet

$$\sum_{i,j} K_{ij} \langle V_j, U_i \rangle = n, \quad \|\text{diag}(\sqrt{p}) K \text{diag}(\sqrt{p})\|_1 = 1, \quad p_i = 1/n.$$

A Schur factorization controls the trace norm of a matrix, not its unrestricted entry sum. The latter pairs the matrix with J_n , whose operator norm is n . This counterexample refutes the inference from the stated invariant. It is not a physical adaptive strategy that refutes the forrelation lower bound.

A valid adaptive extension needs an additional physical contraction retaining the causal measurement structure, or a different adaptive bound with controlled overhead. The saved multiplier-one artifact only inserts the number one after checking the ledger; it does not prove this missing inequality. Its old verdict is historical and is excluded from the supported verification.

The parallel result also does not cover cross-number coherence, a mean-dose budget, or quantum memory between encounters. It does not prove that six is the optimal multipass dose. Run `makecheck` for the supported arithmetic and structural checks, and `makebuild` for the two proof PDFs. The root `AUDIT.md` records exactly what was changed and checked.

References

- [1] Scott Aaronson and Andris Ambainis. Forrelation: A problem that optimally separates quantum from classical computing. In *Proceedings of the Forty-Seventh Annual ACM Symposium on Theory of Computing*, pages 307–316, 2015.